Originality report

COURSE NAME
RDS2S1G5

STUDENT NAME
RUI JIE TANG

FILE NAME
Group07_Final_Report.docx

REPORT CREATED
Aug 28, 2026

Summary

Flagged passages	12	1%
Cited/quoted passages	0	0%

Web matches

roboflow.com	4	0.4%
geeksforgeeks.org	1	0.1%
mathworks.com	1	0.1%
cloud.google.com	1	0.1%
frontiersin.org	1	0.1%
arxiv.org	1	0.1%
linkedin.com	1	0.1%
researchgate.net	1	0.1%
ultralytics.com	1	0.1%

1 of 12 passages

Student passage FLAGGED

Artificial Intelligence (AI) is an important field **of** computer science **that** enables **computers to perform tasks that** normally **require human intelligence, such as** recognition, analysis, **and**

Top web match

Artificial intelligence (AI) is a set **of** technologies **that** empowers **computers to** learn, reason, and **perform** a variety of advanced **tasks** in ways **that** used to **require human intelligence, such as**…

What is Artificial Intelligence (AI)? | Google Cloud https://cloud.google.com/learn/what-is-artificial-intelligence

2 of 12 passages

Student passage FLAGGED

Unlike image classification, which only determines the class of **an entire image, object detection** can identify the categories of **objects and** determine **their locations**

Top web match

Unlike image classification, which assigns a single label to **an entire image, object detection** identifies multiple **objects and their locations** using bounding boxes. Key Concepts in Object Detection.…

Introduction to Object Detection Using Image Processing https://www.geeksforgeeks.org/deep-learning/introduction-to-object-detection-using-image-processing/

---

3 of 12 passages

Student passage FLAGGED

The model uses depthwise separable convolution, **inverted residual blocks**, and **linear bottlenecks to reduce** the number of parameters and **computational** operations **while maintaining**

Top web match

MobileNetV2 (36) is a lightweight convolutional neural network architecture designed for mobile and embedded vision applications. **The** architecture employs **inverted residual blocks** with **linear**…

Lightweight deep learning for medical imaging using MobileNetV2 ... https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2026.1810860/full

---

4 of 12 passages

Student passage FLAGGED

As shown in Figure 3.1, **RetinaNet** is composed **of a ResNet**-50 **backbone**, a Feature Pyramid Network (FPN), **a classification subnet, and a box regression subnet**

Top web match

RetinaNet forms a single FCN comprised **of a ResNet**-FPN **backbone, a classification subnet, and a box regression subnet**, see Figure 3.

arXiv:1708.02002v2 [cs.CV] 7 Feb 2018 https://arxiv.org/pdf/1708.02002

---

5 of 12 passages

Student passage FLAGGED

…between these two measures. IoU is used to evaluate **the overlap between the predicted and ground-truth bounding boxes**. The evaluation results are then used to assess the…

Top web match

What does mAP@0.5 tell you? mAP@0.5 tells us “How well does **the** model detect objects when we require at least 50% **overlap between the predicted and ground-truth bounding boxes**?”

mAP@0.5 vs mAP@0.5:0.95: One IoU Threshold vs Ten https://blog.roboflow.com/map-0-5-vs-map-0-5-0-95/

---

6 of 12 passages

Student passage FLAGGED

A prediction is considered **a** True Positive (TP) when the **predicted bounding box** has **an IoU of at least 0.5 with** a **ground-truth bounding box** and the predicted class is correct. A False Positive…

Top web match

At an IoU threshold of 0.5, **a predicted bounding box** must have **an IoU of at least 0.5 with** the corresponding **ground-truth bounding box** to be considered a True Positive (TP).

mAP@0.5 vs mAP@0.5:0.95: One IoU Threshold vs Ten https://blog.roboflow.com/map-0-5-vs-map-0-5-0-95/

---

7 of 12 passages

Student passage FLAGGED

…ground-truth bounding box and the predicted class is correct. **A False Positive (FP) occurs when a prediction does not correctly match a ground-truth object** or predicts the wrong **class**.

Top web match

A false positive (FP) occurs when the model predicts an object as belonging to **a** class, but the **prediction does not correctly match a ground-truth object** of that **class**.

mAP@0.5 vs mAP@0.5:0.95: One IoU Threshold vs Ten https://blog.roboflow.com/map-0-5-vs-map-0-5-0-95/

---

8 of 12 passages

Student passage FLAGGED

Intersection over Union (IoU) measures **the** degree of **overlap between a predicted bounding box and** its corresponding ground-truth **bounding box**. It is used to evaluate the localisation accuracy of…

Top web match

The **intersection over union (IoU**) metric in object detection quantifies **the overlap between a predicted bounding box and** a ground truth **bounding box**.

Evaluate Object Detector Performance - MATLAB & Simulink https://www.mathworks.com/help/vision/ug/evaluate-object-detector-performance.html

---

9 of 12 passages

Student passage FLAGGED

…predicted bounding box and its corresponding ground-truth bounding box. **It is used to evaluate the** localisation **accuracy of** the **object detection model**.

Top web match

The formula is: **It is** commonly **used to evaluate the** localization **accuracy of** an **object detection model**.

mAP@0.5 vs mAP@0.5:0.95: One IoU Threshold vs Ten https://blog.roboflow.com/map-0-5-vs-map-0-5-0-95/

---

10 of 12 passages

Student passage FLAGGED

A higher F1-score indicates that **the model** achieves **a better balance between Precision and Recall**. A low F1-score may indicate that the model either…

Top web match

A higher F1 Score means **the model** has **a better balance between precision and recall**. Compared to accuracy, which can be misleading on imbalanced datasets, the F1 Score focuses specifically on the…

F1 Score: Balancing Precision and Recall in Classification Models https://www.linkedin.com/posts/bruceratner_f1-score-precision-and-recall-f1-activity-7474228339197714432-85ws

---

11 of 12 passages

Student passage FLAGGED

mAP50-95 **is the** mean AP averaged over ten **IoU thresholds from 0.50 to 0.95 in steps of 0.05**. It is stricter because the higher thresholds require more…

Top web match

mAP@50-95: This **is the** average of mAP calculated at **IoU thresholds from 0.50 to 0.95 in steps of 0.05**. This rigorous metric rewards models that achieve high ...

What is Mean Average Precision (mAP)? - Ultralytics https://www.ultralytics.com/glossary/mean-average-precision-map

---

12 of 12 passages

Student passage FLAGGED

Figure 4.1 also **shows the training and validation loss during** model **training**. The training loss generally decreased as the number of…

Top web match

Loss Curve: This **figure shows the training and validation loss during** the **training** of the four deep learning models (bonnet localization model, blow hole model, head localization model, and whale…

Loss Curve: This figure shows the training and validation loss during... https://www.researchgate.net/figure/Loss-Curve-This-figure-shows-the-training-and-validation-loss-during-the-training-of-the_fig1_312475302
